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WirePlumber SPA Libraries Configuration

A comprehensive guide to SPA (Simple Plugin API) libraries for WirePlumber audio/video session management.

```

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<strong>ALSA (Advanced Linux Sound Architecture)</strong>
```

Purpose: Native Linux kernel audio interface providing PCM (Pulse Code Modulation) playback and capture.

Configuration:

```
context.spa-libs = {  
    api.alsa.* = alsa/libspa-alsa  
}
```

Installation:

Arch Linux

```
sudo pacman -S pipewire alsa-lib
```

Verify installation

```
ls /usr/lib/spa-0.2/alsa/libspa-alsa.so  
spa-inspect /usr/lib/spa-0.2/alsa/libspa-alsa.so
```

Factory Names: api.alsa.enum.udev, api.alsa.pcm.sink, api.alsa.pcm.source

ALSA SPA Reference

BlueZ5 (Bluetooth Audio)

Purpose: Bluetooth audio device support with A2DP, HSP, HFP codec support.

Configuration:

```
context.spa-libs = {  
    api.bluez5.* = bluez5/libspa-bluez5  
}
```

Installation:

Arch Linux

```
sudo pacman -S pipewire-pulse bluez bluez-libs
```

Enable Bluetooth service

```
systemctl --user enable --now bluetooth.service
```

Verify installation

```
ls /usr/lib/spa-0.2/bluez5/libspa-bluez5.so
```

Factory Names: api.bluez5.enum.dbus, api.bluez5.midi.enum

BlueZ5 WirePlumber Reference

libcamera (Modern Camera API)

Purpose: Modern Linux camera framework supporting complex camera hardware with advanced features.

Configuration:

```
context.spa-libs = {  
api.libcamera.* = libcamera/libspa-libcamera  
}
```

Installation:

bash

Arch Linux

sudo pacman -S libcamera pipewire

Verify installation

ls /usr/lib/spa-0.2/libcamera/libspa-libcamera.so

Factory Names: api.libcamera.enum.client, api.libcamera.source

libcamera Official Site

V4L2 (Video4Linux2)

Purpose: Legacy video capture API for webcams and video devices.

Configuration:

```
context.spa-libs = {  
api.v4l2.* = v4l2/libspa-v4l2  
}
```

Installation:

bash

Arch Linux (usually included by default)

sudo pacman -S v4l-utils pipewire

List video devices

v4l2-ctl -list-devices

Verify installation

ls /usr/lib/spa-0.2/v4l2/libspa-v4l2.so

Factory Names: api.v4l2.source, api.v4l2.enum.udev

V4L2 Kernel Documentation

Vulkan (Graphics Compute API)

Purpose: GPU-accelerated video/image processing using Vulkan compute shaders.

Configuration:

```
lua context.spa-libs = { api.vulkan.* = vulkan/libspa-vulkan }
```

Installation:

```
bash
```

Arch Linux

```
sudo pacman -S vulkan-icd-loader vulkan-tools pipewire
```

Verify Vulkan support

```
vulkaninfo | head -20
```

Verify installation

```
ls /usr/lib/spa-0.2/vulkan/libspa-vulkan.so
```

Factory Names: api.vulkan.compute.source

Vulkan Official Site

AEC (Acoustic Echo Cancellation)

Purpose: Real-time acoustic echo cancellation for voice calls and conferencing.

Configuration:

```
lua context.spa-libs = { audio.aec.* = aec/libspa-aec }
```

Installation:

```
bash
```

Arch Linux (requires WebRTC audio processing)

```
sudo pacman -S pipewire webrtc-audio-processing
```

Verify installation

```
ls /usr/lib/spa-0.2/aec/libspa-aec.so spa-inspect /usr/lib/spa-0.2/aec/libspa-aec.so
```

Factory Names: audio.aec.webrtc

Echo Cancellation Module Documentation

Audio Convert (Format Conversion)

Purpose: Audio format, sample rate, and channel conversion with resampling.

Configuration:

```
lua context.spa-libs = { audio.convert.* = audioconvert/libspa-audioconvert }
```

Installation:

```
bash
```

Arch Linux (core component)

```
sudo pacman -S pipewire
```

Verify installation

```
ls /usr/lib/spa-0.2/audioconvert/libspa-audioconvert.so
```

Factory Names: audio.convert, audio.resample, audio.channelmix

Audio Convert SPA Reference

Audio Mixer (Stream Mixing)

Purpose: Combines multiple audio input streams into a single output stream.

Configuration:

```
context.spa-libs = {  
audio.mixer.* = audiomixer/libspa-audiomixer  
}
```

Installation:

```
sudo pacman -S pipewire
```

```
ls /usr/lib/spa-0.2/audiomixer/libspa-audiomixer.so
```

Factory Names: audio.mixer.dsp

Audio Mixer SPA Reference

Control (Device Control)

Purpose: Control signal mixer for device control streams.

Configuration:

```
context.spa-libs = {  
  control.* = control/libspa-control  
}
```

Installation:

```
bash
```

Arch Linux (core component)

```
sudo pacman -S pipewire
```

Verify installation

```
ls /usr/lib/spa-0.2/control/libspa-control.so
```

Factory Names: control.mixer

Control SPA Reference

Support (Core Infrastructure)

Purpose: Essential core support including CPU detection, logging, event loops, and system utilities.

Configuration:

```
context.spa-libs = {  
  support.* = support/libspa-support  
}
```

Installation:

```
# Arch Linux (core component, always required)
```

```
sudo pacman -S pipewire
```

```
# Verify installation
```

```
ls /usr/lib/spa-0.2/support/libspa-support.so  
spa-inspect /usr/lib/spa-0.2/support/libspa-support.so
```

Factory Names: support.cpu, support.logger, support.loop, support.system

Support SPA Reference

Video Convert (Video Format Conversion)

Purpose: Video format, resolution, and colorspace conversion.

Configuration:

```
lua context.spa-libs = { video.convert.* = videoconvert/libspa-videoconvert }
```

Installation:

```
# Arch Linux (core component)
```

```
sudo pacman -S pipewire
```

```
# Verify installation
```

```
ls /usr/lib/spa-0.2/videoconvert/libspa-videoconvert.so
```

Factory Names: video.convert, video.scale

Video Convert SPA Reference